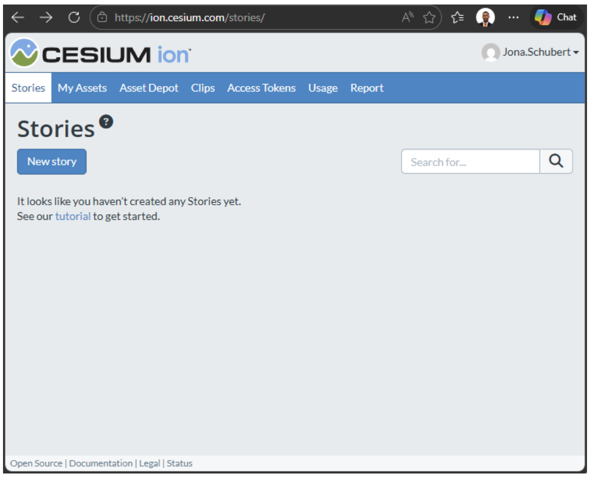
Digital Twin Workshop – Student Guide

This guide supports the Digital Twin workshop and is designed to help you follow along, reflect on what you are building, and understand how each step contributes to a real digital twin workflow.

0. Creating a Cesium account and accessing workshop assets

Before we start building anything, everyone needs to be working from the same starting point.

Create a Cesium ion account by visiting [Cesium: The Platform for 3D Geospatial] and signing up with your university or personal email address. Once logged in, you will arrive at the Cesium ion dashboard.



This dashboard is where you will access the tools used in today's workshop, including Cesium Stories, Cesium Sandcastle, assets, and access tokens.

***No local installation is required

To save time during the session, the assets and dataset(s) used in this workshop are provided via the instructor's GitHub portfolio. These include boundaries, pre-tiled photogrammetry, and example models.

1. Workshop mindset

You are not expected to memorise code or understand every line you see today. The goal is to understand patterns, not syntax.

As you work through the labs, keep coming back to these three questions:

- What data is being used?
- What question does it help answer?
- How could this scale up to something bigger?

2. What you are building today

Throughout the workshop you will incrementally build a simple digital twin. Not a finished product, but a realistic foundation.

By the end, your digital twin will include:

- Real-world terrain and global context
- Existing buildings and captured reality
- A proposed design model
- Tabular data connected to geometry
- Basic interaction and controls

3. Cesium Sandcastle

Cesium Sandcastle is a live coding environment for CesiumJS. It runs real production code directly in the browser. Why we use Sandcastle in this workshop:

- No installation or setup
- Instant feedback
- Same code used in real applications
- Safe environment for experimentation

Sandcastle is not the final destination, but the skills and patterns you learn here transfer directly to real projects.

4. Cesium ion assets (quick reference)

In Cesium ion, everything is an asset. An asset is cloud-hosted, optimised, and streamable data.

Common asset types you will see today:

• Terrain and imagery
• 3D Tiles (buildings, photogrammetry, BIM models)
• Vector data such as GeoJSON

Once uploaded, assets are referenced by ID and can be reused across viewers, apps, and stories.

5. Access tokens (quick reference)

Access tokens authenticate your application with Cesium ion.

Key ideas to remember:

• Tokens control access, not the data itself
• Tokens have scopes (permissions)
• Best practice is one token per application

In this workshop, tokens are provided so you can focus on learning, not configuration.

6. Vibecoding / prompt engineering

Alongside Cesium, you will use Large Language Models (LLMs) to help you build faster.

This works best when you give the model clear guardrails:

• Describe what you want to build
• Specify constraints (Sandcastle, CesiumJS)
• Be clear about data sources and goals

Prompt engineering is simply learning how to communicate intent clearly.

7. Lab structure

The labs are structured to mirror how real projects evolve over time:

Lab 1 – Establishing context (area of interest, existing buildings)
Lab 2 – Captured reality (photogrammetry) or Googe photorealistic 3D tiles
Lab 3 – Design intent (proposed model)
Lab 4 – Connecting data (turning models into insight)

Each lab layers new information on top of what already exists.

8. Prompt cheat sheet

A prompt cheat sheet is provided alongside this guide. Use it to:

- Set context for the LLM
- Ask for CesiumJS/Sandcastle-compatible code
- Keep generated code modular and understandable

10. What success looks like

Success today is not finishing every task.

Success is understanding how terrain, assets, data, and interaction come together to form a digital twin — and knowing how to grow it further.